

Nigel Chao

Game QA & Software Engineer
Gothenburg, Sweden

Mobile: (+46) 702 911 623
Portfolio: niguelchaos.github.io
E-mail: niguelchaos@gmail.com
Linkedin: linkedin.com/in/nigel-chao/

SUMMARY

- 2+ years of experience in **test automation and quality process improvement** with academic training in game design.
- Hands-on development experience with critical evaluation skills to analyze gameplay systems, UX, and market potential with gameplay deconstruction and mechanics analysis.
- Global experience across Asia and Europe - polyglot in English, Mandarin, Cantonese, Portuguese, basic Swedish.

PROFESSIONAL EXPERIENCE

Software Engineer | Repli5

2022 - 2024

- Developed 15+ features using Python for WorldGenerator, a Blender add-on that automates the creation of 3D environments by working in diverse domains such as Testing, UI, Technical Writing, and DevOps.
- Designed and implemented unit tests for critical features to validate functionality and ensure code reliability, utilizing mocking techniques to isolate them.
- Drove over 100+ Scrum ceremonies, fostering a culture of collaboration and continuous improvement.
- Wrote Python scripts to automate regression testing for Blender plugin UI, saving 10+ hours/week.
- Streamlined build and deployment processes by scripting automation for various build stages, including automating the execution of unit tests and deployment across environments using Google Cloud and Bitbucket Pipelines.
- Migrated QA & internal user documentation to Notion, enabling self-service access for artists and PMs.

PERSONAL PROJECTS

Game Designer | “Scarab Descent”

2024

- Designed and greyboxed a **non-linear world map** in UE5 using a modular kit, emphasizing genre staples such as exploration and ability gated progression for a single player Metroidvania FPS vertical slice.
- Drafted design documents outlining enemy placement and mechanics enhancing combat and pacing of player flow.
- Scripted AI, combat and traversal mechanics using Blueprints and GAS (Gameplay Ability System).

EDUCATION

Master’s in Computer Science, Major in Game & Interaction Design

University of Gothenburg | Graduated 2022

- AR Mobile Game (Unity): 2-Player physics game, explode enemy fortresses made of blocks. Performed exploratory and end to end testing to ensure 10+ critical features were functional and performant.

Bachelor’s in Computer Science, Major in Software Engineering & Management

University of Gothenburg | Graduated 2020

- Cyber-Physical Systems (C++): Docker microservices for a mini-vehicle to handle traffic situations. Performed **local, recorded, and onsite/system testing** to ensure critical software features performed **on the embedded hardware and on the designated site**.

SKILLS

- Game & System Design
- Gameplay Deconstruction & Feature Analysis
- Test Plan Development
- Automated Testing & Bug Reporting
- Technical Writing & Documentation

TECHNOLOGIES

- **Programming:** Python, C#, C/C++, Java, JavaScript, Bash, Blueprints
- **Version Control:** Git, Perforce, Plastic SCM
- **Collaboration Tools:** Miro, Figma, Jira, Notion, Trello
- **Game Engines:** Unity, Unreal Engine
- **Containerization:** Docker